

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main() { int n; if (scanf("%d", &n) != 1) return 0;
long long *a = (long long *)malloc(n * sizeof(long long));
for (int k = 0; k < n; k++) {
    scanf("%lld", &a[k]);
}
int i = 0;
int j = n - 1;
while (i < n && j >= 0) {
    if(a[i]>a[j]) {
        printf("1 ");
        j--;
    } else if (a[i] < a[j]) {
        printf("2 ");
        i++;
    } else {
        printf("0 ");
        i++;
        j--;
    }
}
printf("\n");
free(a);
return 0;
}
```

Input:

```
3
1 2 3
```

Output:

```
2 2 0
```